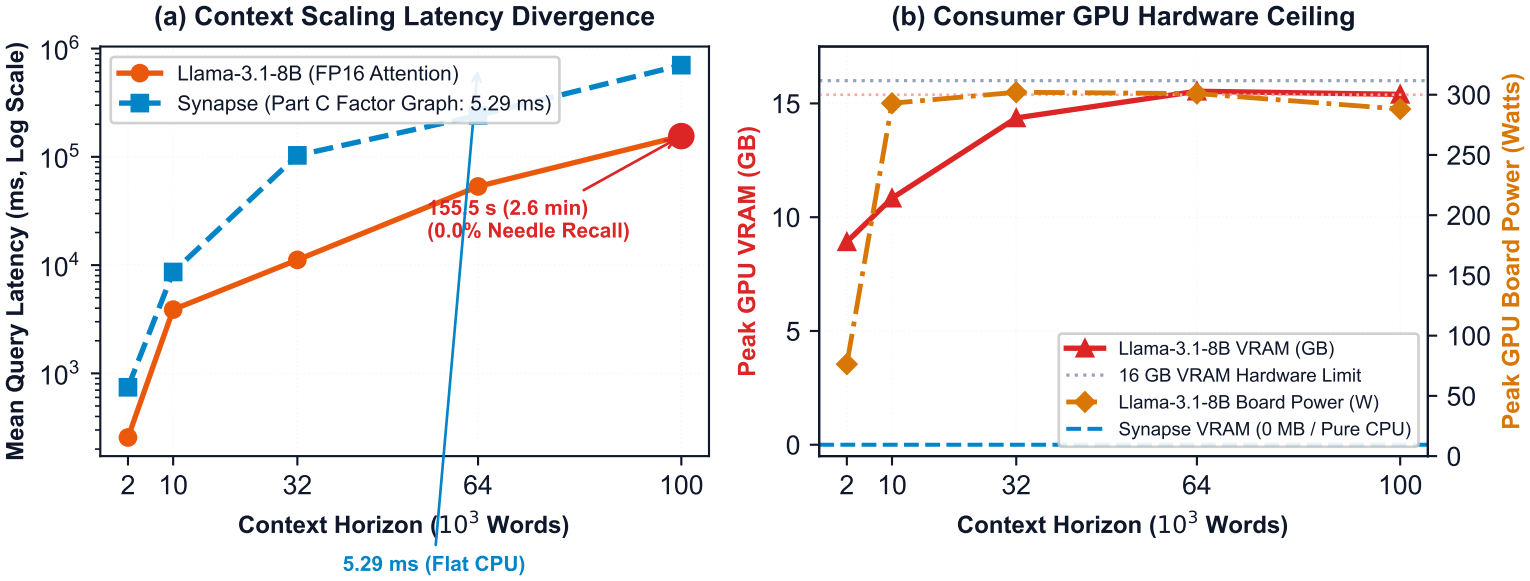
(a) Context Scaling Latency Divergence
Mean Query Latency (ms, Log Scale)
10^6
10^5
10^4
10^3
Llama-3.1-8B (FP16 Attention)
Synapse (Part C Factor Graph: 5.29 ms)
155.5 s (2.6 min)
(0.0% Needle Recall)
5.29 ms (Flat CPU)
2
10
32
64
100
Context Horizon (10^3 Words)
(b) Consumer GPU Hardware Ceiling
Peak GPU VRAM (GB)
Peak GPU Board Power (Watts)
0
5
10
15
0
50
100
150
200
250
300
Llama-3.1-8B VRAM (GB)
16 GB VRAM Hardware Limit
Llama-3.1-8B Board Power (W)
Synapse VRAM (0 MB / Pure CPU)
2
10
32
64
100
Context Horizon (10^3 Words)